THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. From a solid cylinder of height 48 cm and base radius 14 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

(a) 5456 cm² **(b)** 7040 cm² **(c)** 6424 cm² **(d)** 7656 cm²

2. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50 and frequencies 8, 16, p, 10, 8 respectively is 23.8. The value of p is: 1

(a) 13 **(b)** 8 **(c)** 10 **(d)** 7

3. A box contains 25 tickets numbered 1 to 25. One ticket is drawn at random. The probability that the number drawn is a perfect square is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{6}{25}$ **(c)** $\frac{4}{25}$ **(d)** $\frac{4}{5}$

4. A wooden article is made by scooping out a hemisphere of radius 7 cm from each end of a solid cylinder of height 15 cm and base radius 7 cm. The total surface area of the article is: 1

(a) 660 cm² **(b)** 968 cm² **(c)** 1276 cm² **(d)** 1584 cm²

5. In the formula $\bar{x} = a + h\dfrac{\sum f_i u_i}{\sum f_i}$ (step-deviation method), u_i equals: 1

(a) $\dfrac{x_i - a}{h}$ **(b)** $h(x_i - a)$ **(c)** $\dfrac{x_i}{h}$ **(d)** $x_i - a$

6. A lot of 90 ball pens contains 10 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{8}{9}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{9}{10}$ **(d)** $\frac{79}{90}$

7. A solid metallic cylinder of base radius 6 cm and height 24 cm is melted and recast into solid spheres of radius 2 cm. The number of spheres obtained is: 1

(a) 87 **(b)** 79 **(c)** 81 **(d)** 162

8. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 13, 5, 20, 10, 2 respectively, The mean of the data is: 1

(a) 216 **(b)** 22.6 **(c)** 21.6 **(d)** 25

9. Two dice are thrown simultaneously. The probability that both dice show the same number (a doublet) is: 1

(a) $\frac{7}{36}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{5}{36}$

10. A solid metallic cone of base radius 3 cm and height 12 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

(a) 6 cm **(b)** 4 cm **(c)** 2 cm **(d)** 3 cm

11. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 10, 11, 10, 13, 6 respectively, The median of the data is: 1

(a) 40 **(b)** 33.8 **(c)** 30 **(d)** 34

12. A coin is tossed three times. The probability of getting all three faces the same is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{3}{8}$

13. A toy is a cone of base radius 35 cm and height 12 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 9020 cm² **(b)** 11770 cm² **(c)** 4070 cm² **(d)** 15620 cm²

14. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 2, 11, 8, 17, 2 respectively, The mode of the data is: 1

(a) 30 **(b)** 26.5 **(c)** 40 **(d)** 33.75

15. A letter is chosen at random from the word 'EXAMINATION'. The probability that it is a consonant is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{6}{11}$ **(d)** $\frac{5}{12}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	60–70	70–80	80–90	90–100	100–110
Number of workers	12	20	3	2	3

Find the mean.

17. A coin is tossed three times. Find the probability of getting a head on the first toss. 2

18. A solid metal cuboid 24 cm by 12 cm by 15 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 2

19. The daily electricity consumption (in units) of households in a colony is given below. 2

Units consumed	50–60	60–70	70–80	80–90	90–100	100–110
Number of households	8	10	14	2	13	3

Find the mode.

20. A red die and a blue die are rolled together. Find the probability that the numbers differ by 2. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A box contains 28 tickets numbered 1 to 28. One ticket is drawn at random. Find the probability that the number drawn is a prime number. 3

22. From a solid cylinder of height 24 cm and base radius 7 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

23. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 3

Lifetime (hours)	0–10	10–20	20–30	30–40	40–50
Number of bulbs	8	16	8	10	8

Find the mode.

24. A bag contains 7 red and 5 blue balls. 5 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 3

OR

A toy is a cone of base radius 10.5 cm and height 14 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

25. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	20–30	30–40	40–50	50–60	60–70
Number of patients	4	13	4	16	3

Find the mean.

OR

Two dice are thrown simultaneously. Find the probability that the first number is greater than the second. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 7,
(ii) the sum is at least 10,
(iii) the sum is less than 5.

OR

A conical vessel of radius 6 cm and height 12 cm is full of water. The water is poured into a cylindrical vessel of base radius 3 cm. Find the height of the water in the cylindrical vessel. 5

27. The median of a distribution with classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 15, 10, 19, x, 16, y, 16 respectively, and total frequency 100, is 71.25. Find the value of x. 5

OR

A bag contains 5 red and 7 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{8}{15}$. Find the value of x. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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